

Nitin Singh Rathore

+1 817 819 8146 | nxr3560@mavs.uta.edu | [linkedin.com/in/nitin-singh-rathore](https://www.linkedin.com/in/nitin-singh-rathore) | github.com/Nitin3560

Technical Skills

Languages: Python, C++ (C++17), Java, JavaScript/TypeScript, Go, SQL

CS Fundamentals: Data Structures & Algorithms, Operating Systems, Networking, Distributed & Parallel Systems, Concurrency

AI & ML Systems: Natural Language Processing, Information Retrieval, RAG, Vector Search, Embeddings, LLM APIs

Backend & APIs: FastAPI, Node.js, REST APIs, GraphQL, Microservices, Event-Driven Workers, Asynchronous Processing

Web & Data: React/Next.js, HTML5/CSS3, PostgreSQL, Redis, Kafka, MySQL, pgvector, Query Optimization

Systems & Tooling: Linux/Unix, AWS, Docker, Kubernetes, Terraform, CI/CD, Git, Code Review, Testing

Work Experience

Software Engineer

Sept 2023 – Oct 2024

WERBOOZ Pvt. Ltd

Indore, India

- Built and maintained 6 production backend services in Java and Apex, delivering REST APIs across 3 client applications, cutting manual processing ~40%
- Refactored 15+ high-latency SQL/SOQL queries across backend microservices, improving query performance by 35% via indexing and caching
- Engineered REST and SOAP API integrations with 5+ internal and third-party systems, with JSON/XML mapping and retry logic
- Troubleshoot and resolved production defects, automating regression and API testing in CI/CD to cut post-release defects by 30%

Software Engineer Intern

Feb 2023 – Sept 2023

WERBOOZ Pvt. Ltd

Indore, India

- Refactored Java service and data-access layers into reusable components, improving repository performance by ~15%
- Delivered 4 features across 2 releases through Git pull requests and peer code review, expanding testing to prevent 20+ defects

Graduate Teaching Assistant

Aug 2025 – Present

CSE Department, University of Texas at Arlington

Arlington, Texas

- Built a Python tool automating grading for 50+ weekly assignments; mentored 100+ students debugging Python and C/C++ across Data Science and Machine Learning coursework

Projects

YoMeets | Real-Time AI Meeting & Workflow Automation Assistant

Jun 2026 – July 2026

TypeScript, Node.js, PostgreSQL/pgvector, Deepgram, LLM APIs, GitHub/Google APIs

- Built a real-time Node.js/TypeScript assistant using NLP extraction and RAG, at 94% precision and <3s p95 latency
- Automated developer workflows across 3 REST API integrations, with 98% success and 0 duplicate side effects

CareerOS | Full-Stack Search & Ranking Platform

July 2026 – Present

Python, FastAPI, React/Next.js, PostgreSQL, Redis/RQ, Docker

- Built a platform end-to-end: React/Next.js frontend, FastAPI services, event-driven workers indexing 31,200+ documents
- Cut median ranking latency from ~690 ms to ~3.5 ms at 32K+ records by profiling a Python bottleneck and moving filtering and ranking into PostgreSQL

CloudQueue | Distributed Task Queue for Web Scraping Jobs

Mar 2026 – Jun 2026

Python, Redis, AWS, Docker, Kubernetes, Terraform, Linux

- Built a REST API-driven queue distributing scraping jobs across a concurrent worker pool at ~1.4K tasks/sec, with retry backoff and dead-lettering
- Implemented at-least-once delivery with idempotent execution, recovering from crashes in <10s with 0 duplicates

Research & Publications

IEEE CSCN 2026: Integrity-Aware Digital Twin Synchronization for ISAC-Enabled UAV Networks

M.S. Thesis: Cross-Layer Supervisory Control for Low-Altitude UAV Swarm Networks

Education

University of Texas at Arlington

M.S. Computer Science | Expected Dec 2026

Arlington, Texas

Acropolis Institute of Technology & Research

B.Tech. Computer Science | 2019 – 2023

India